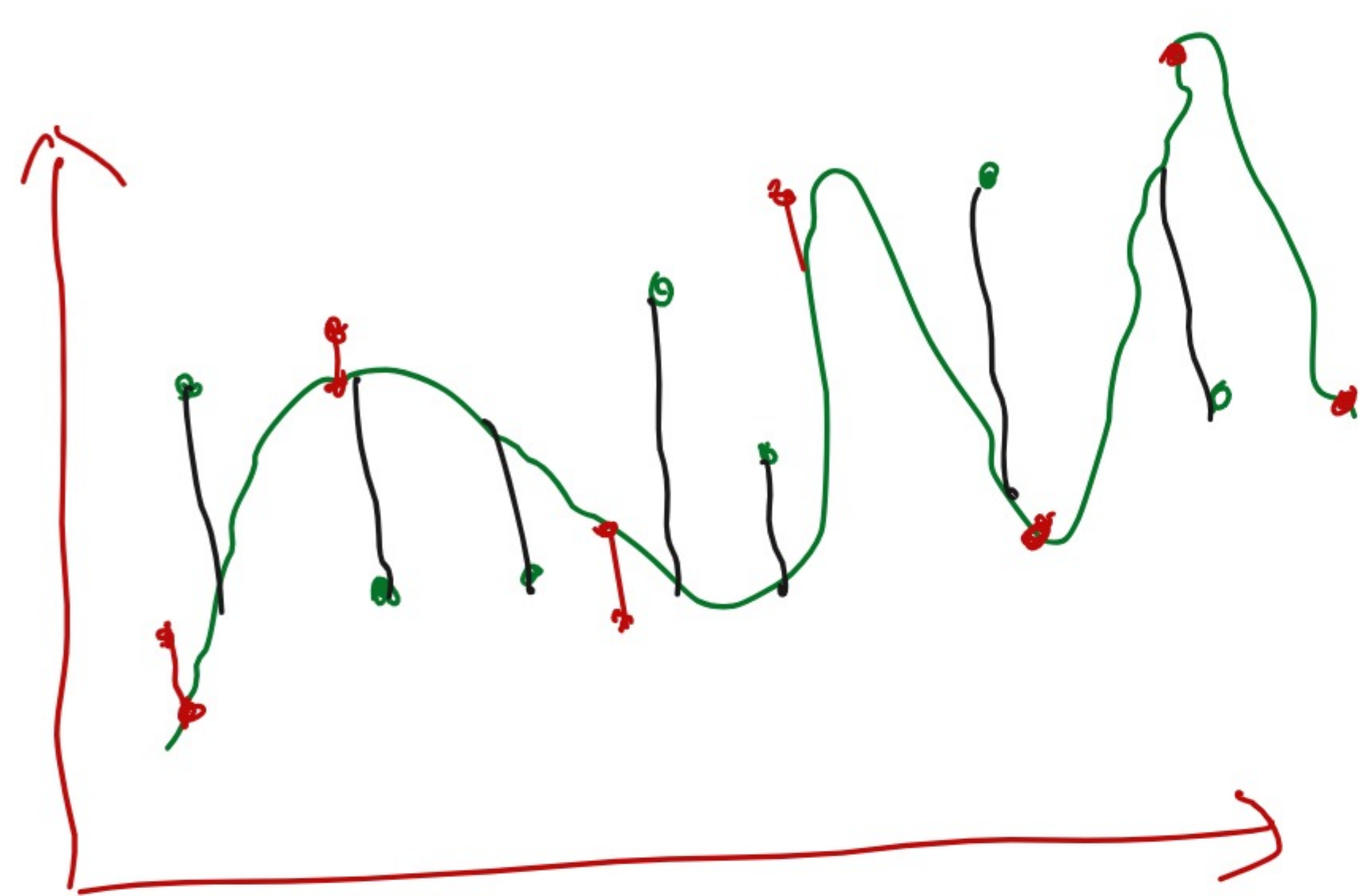
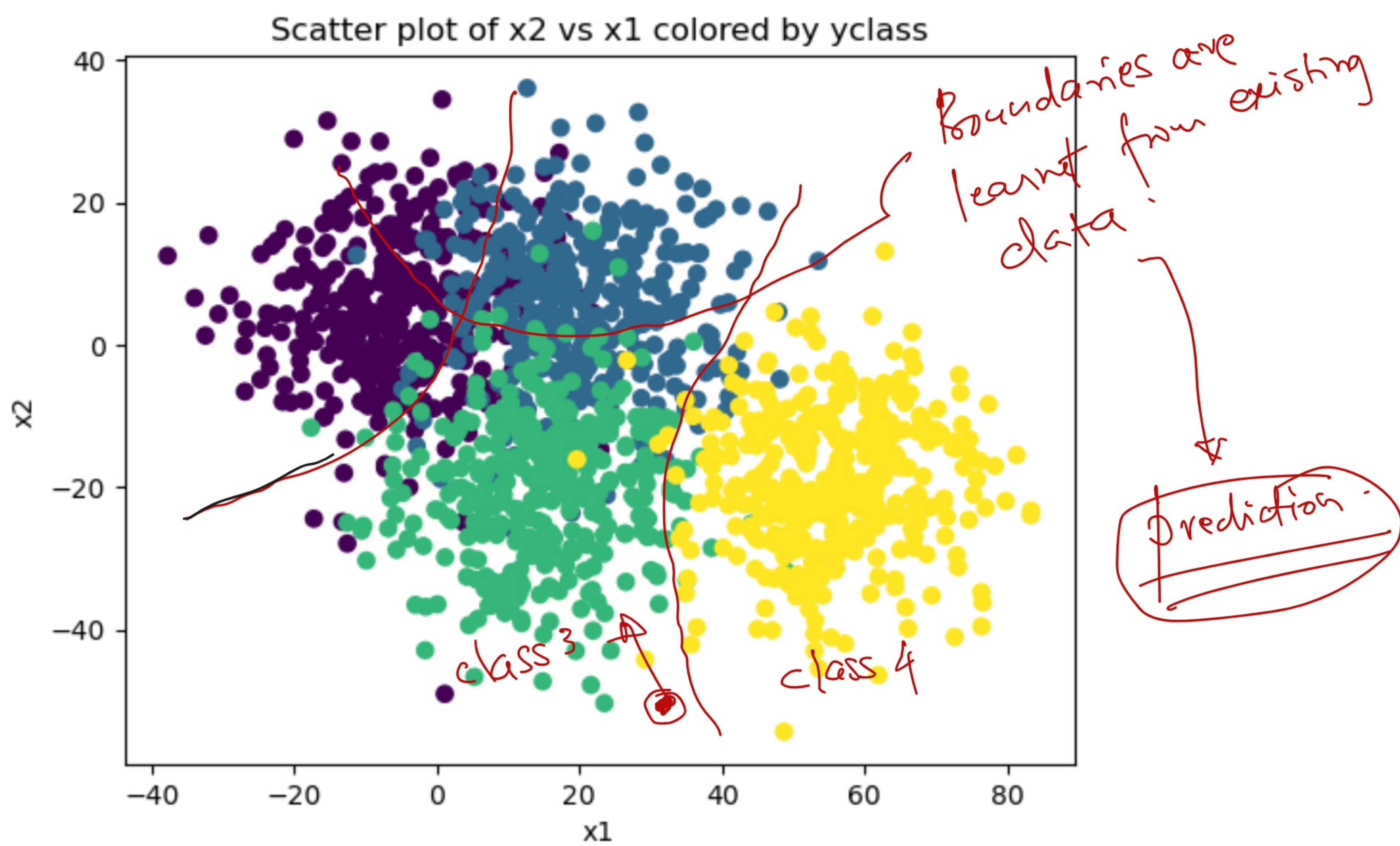
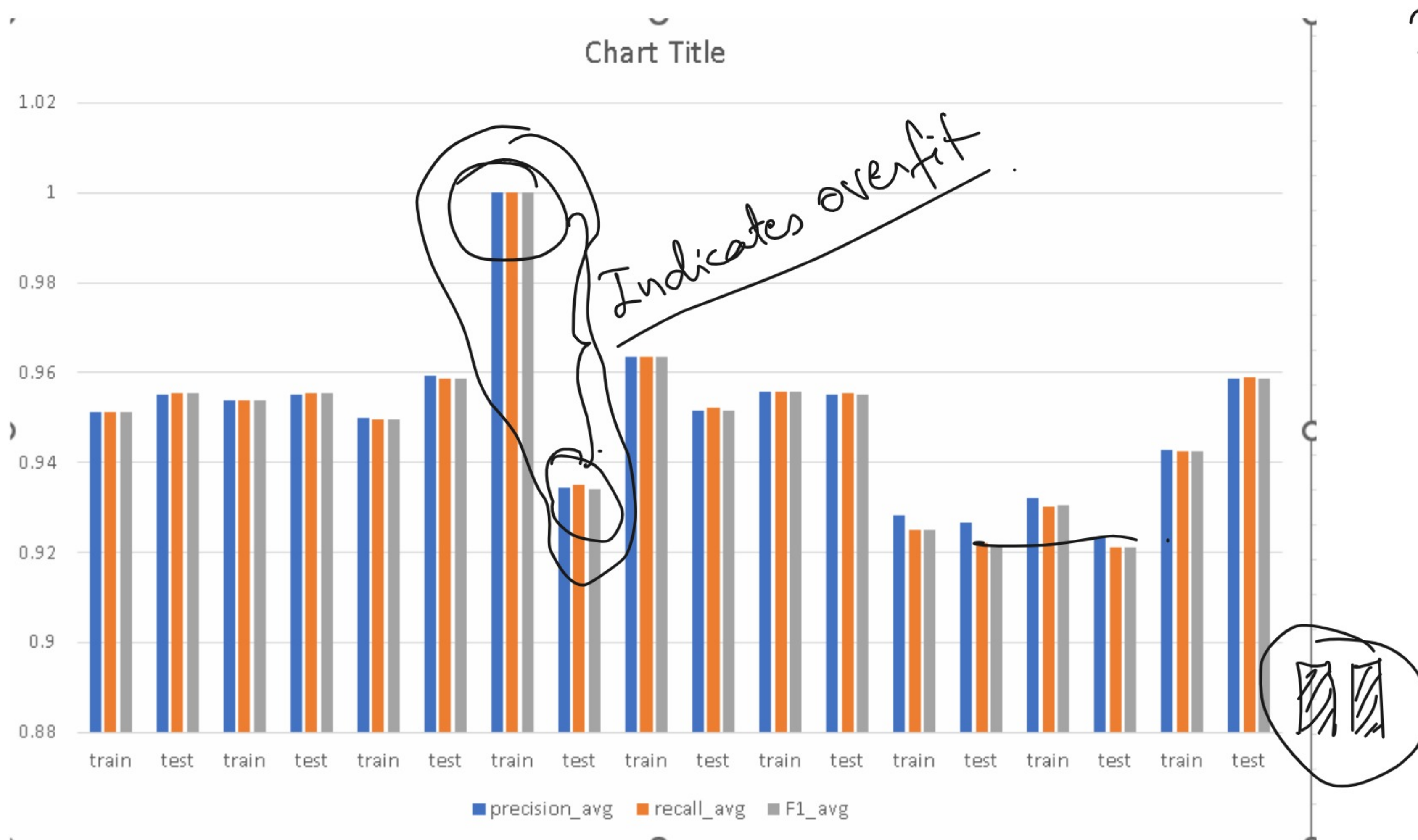




$\sum$ test errors : large value -
 $\sum$ train errors : smaller value

} diff is rel. large



Penalty of 2 marks

"if you do not submit the exercises"

50 → 20

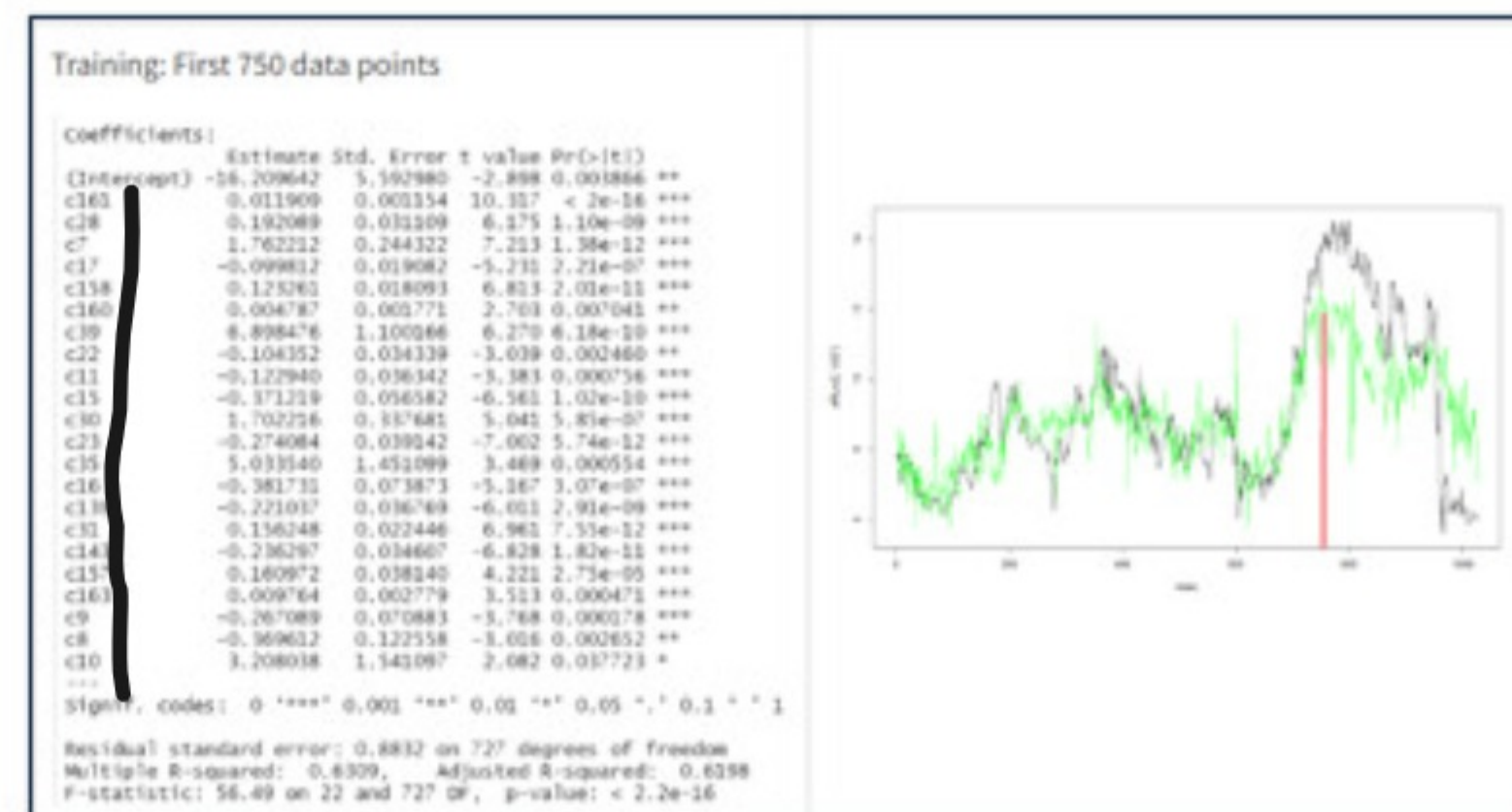
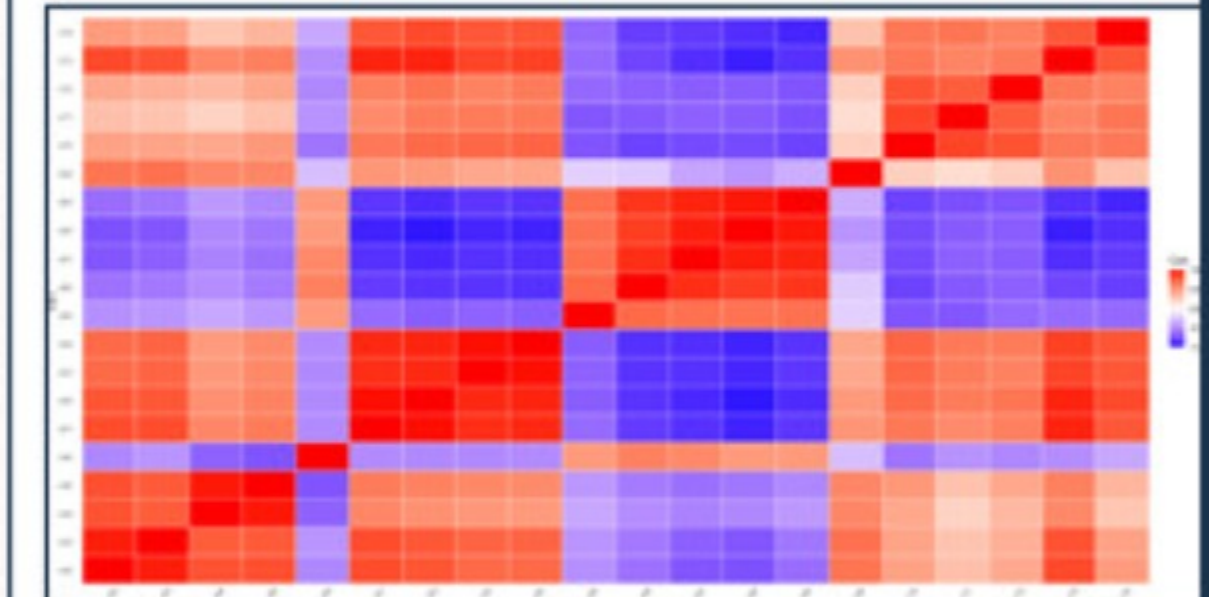
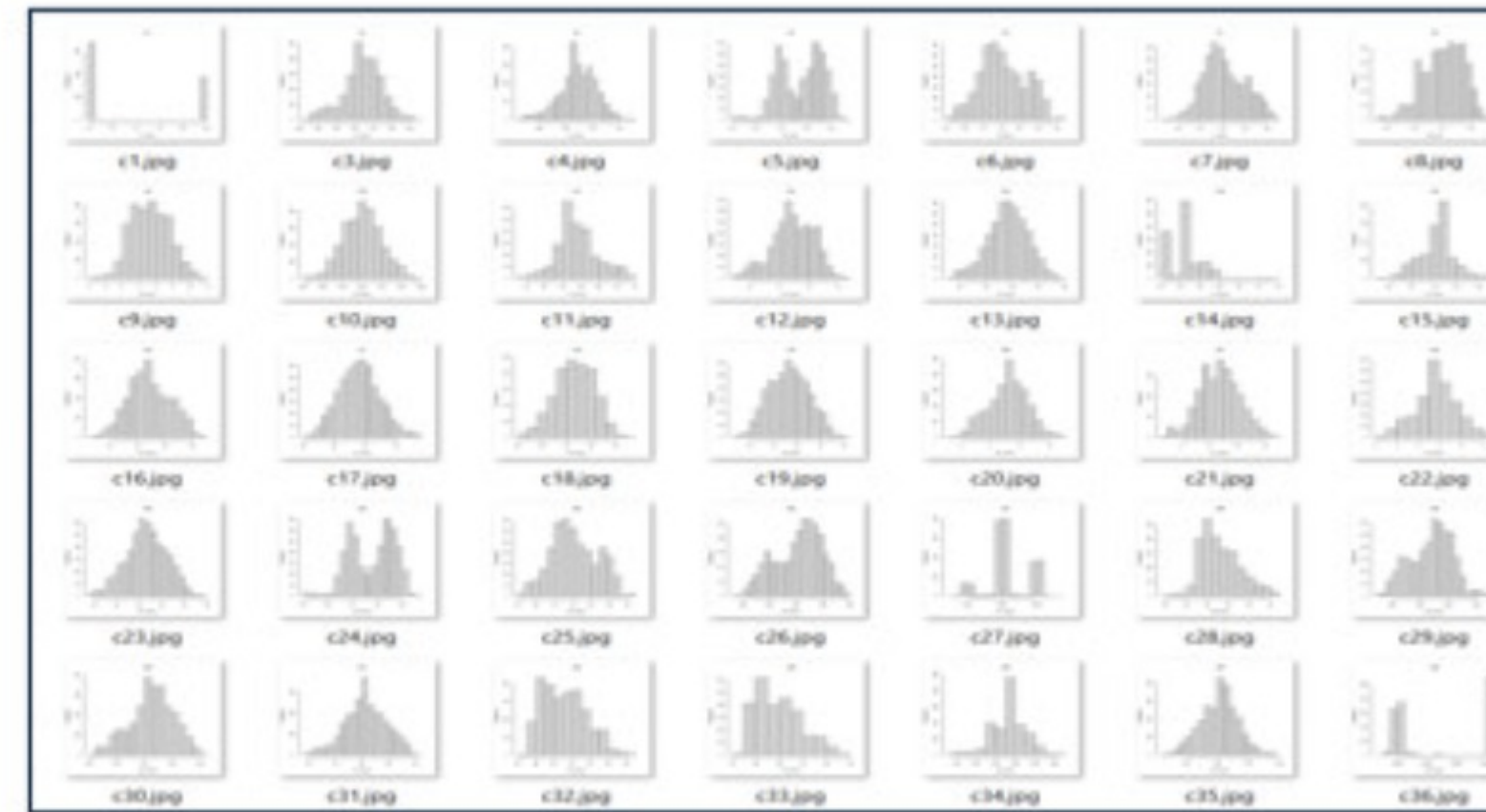
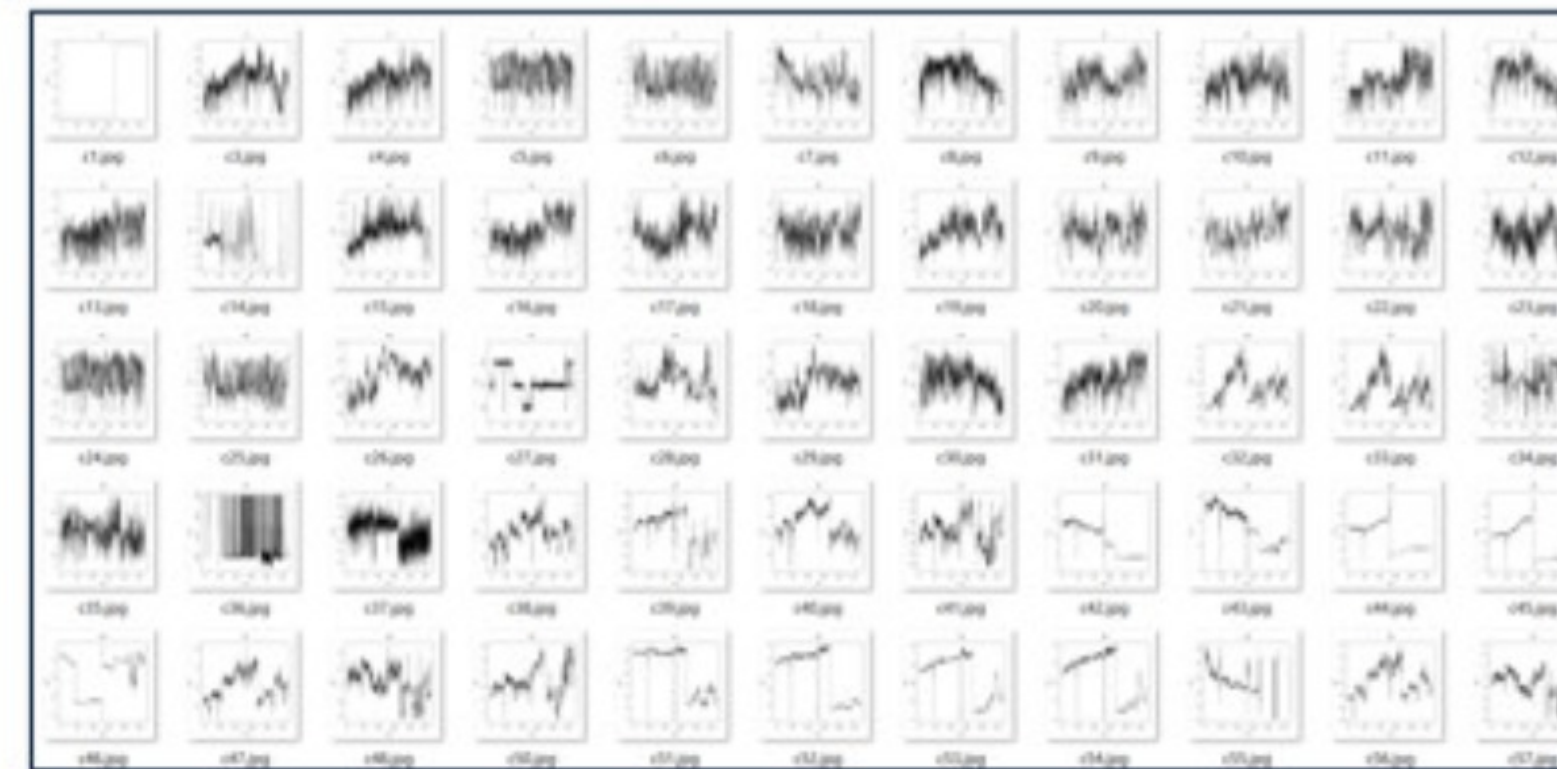
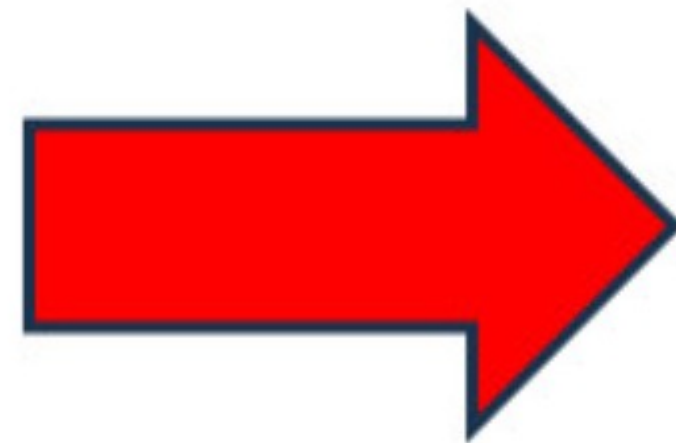
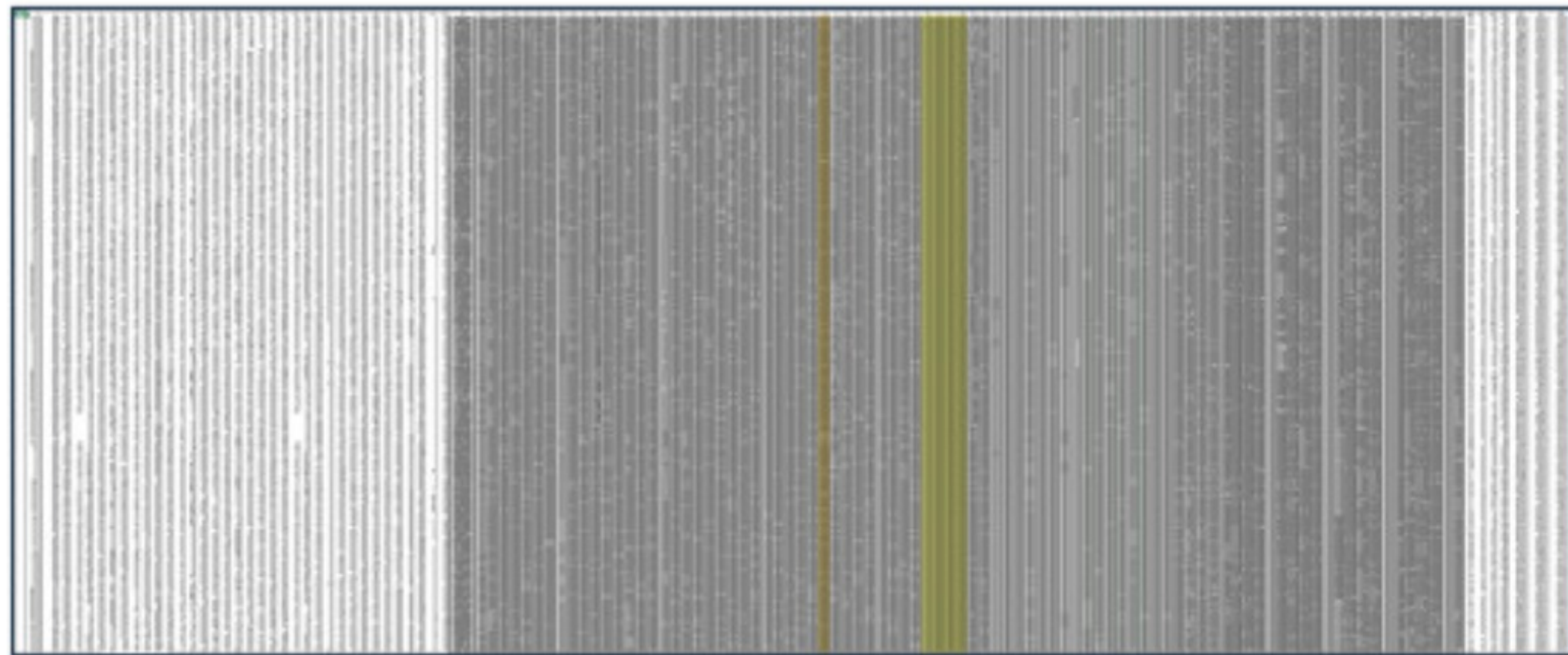


$y = f(x)$

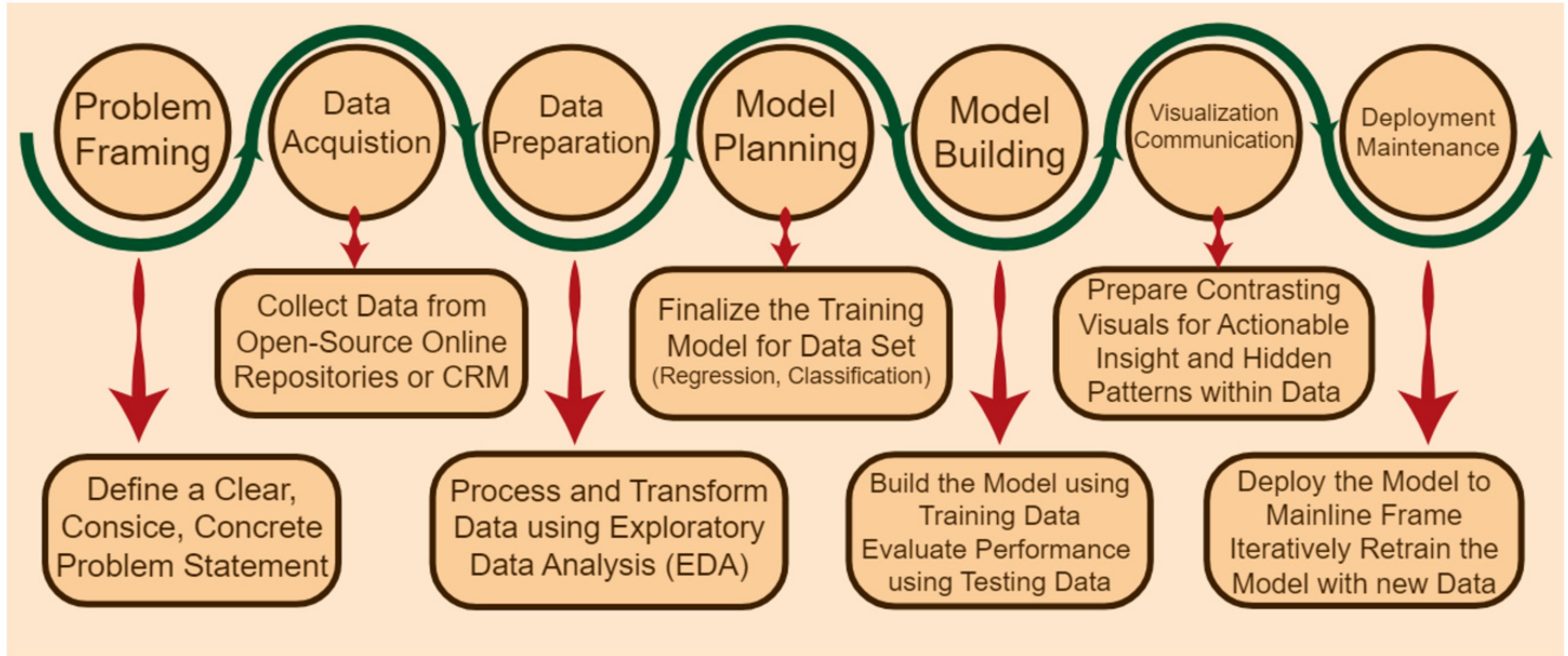
discrete → class

continuous → regression

Goal of Data Science : **Extract Insights from Data**



The Data Science Process



- CRISP-DM
 - Cross Industry Standard Process for Data Mining
- Six steps – run cyclically

p 1. Business Understanding

✓ 2. Data Understanding EDA.

3. Data Preparation ⇒

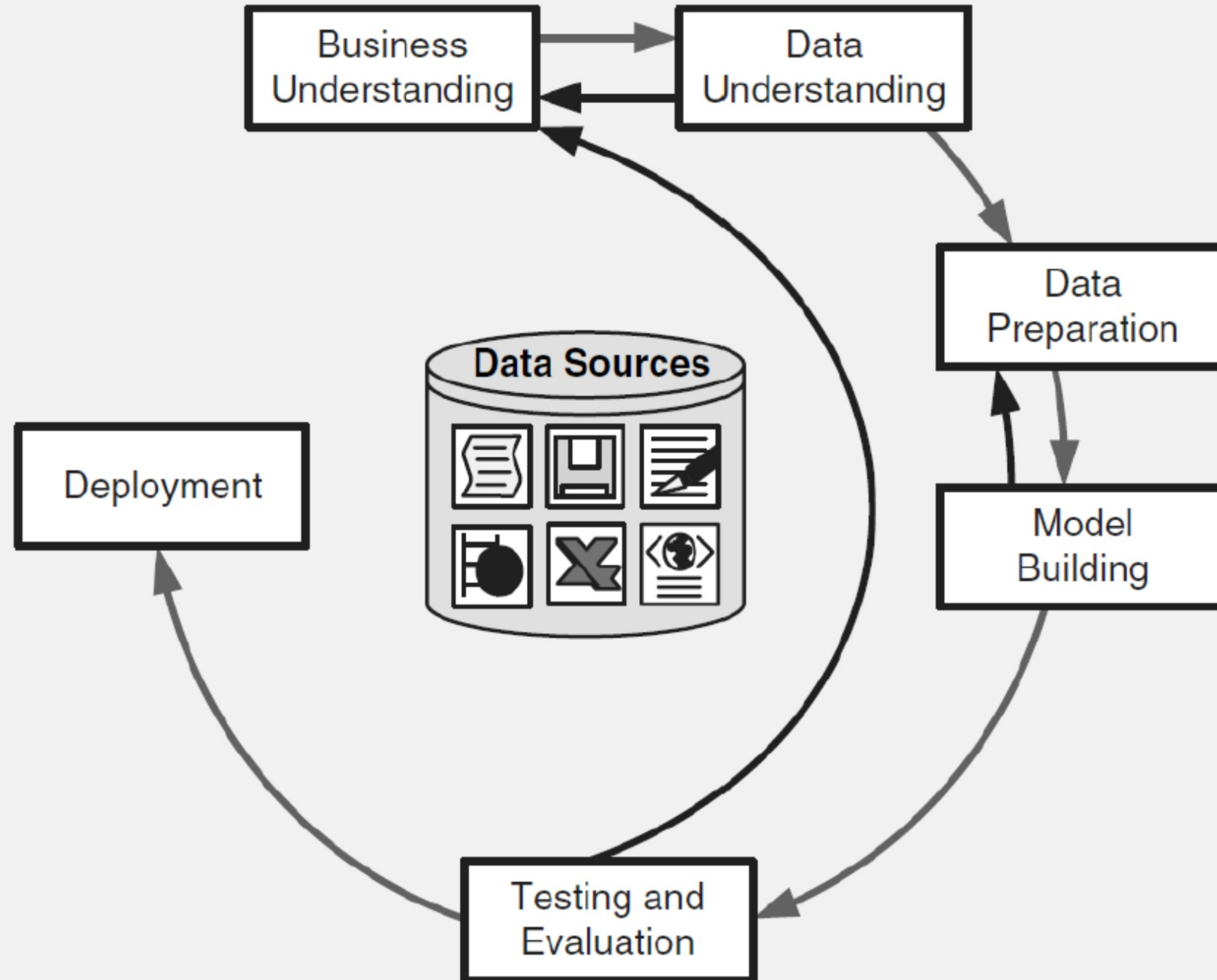
→ 4. Modelling

→ 5. Evaluation

p [6. Deployment]

→ IT, cloud, VM, communication,
Deployment.

} 80% -
=



Business Understanding	Data Understanding	Data Preparation	Modeling	Evaluation	Deployment
Determine Business Objectives Background Business Objectives Business Success Criteria	Collect Initial Data Initial Data Collection Report	Select Data Rationale for Inclusion/Exclusion	Select Modeling Techniques Modeling Technique Modeling Assumptions	Evaluate Results Assessment of Data Mining Results w.r.t. Business Success Criteria Approved Models	Plan Deployment Deployment Plan
Assess Situation Inventory of Resources Requirements, Assumptions, and Constraints Risks and Contingencies Terminology Costs and Benefits	Describe Data Data Description Report	Clean Data Data Cleaning Report	Generate Test Design Test Design	Review Process Review of Process	Plan Monitoring and Maintenance Monitoring and Maintenance Plan
Determine Data Mining Goals Data Mining Goals Data Mining Success Criteria	Explore Data Data Exploration Report	Construct Data Derived Attributes Generated Records	Build Model Parameter Settings Models Model Descriptions	Determine Next Steps List of Possible Actions Decision	Produce Final Report Final Report Final Presentation
Produce Project Plan Project Plan Initial Assessment of Tools and Techniques	Verify Data Quality Data Quality Report	Integrate Data Merged Data	Assess Model Model Assessment Revised Parameter Settings		Review Project Experience Documentation
		Format Data Reformatted Data Dataset Dataset Description			

Exploratory Data Analysis (EDA) is an approach used in statistics and data science to analyze and investigate data sets, with the goal of summarizing their main characteristics and discovering patterns and trends. It involves performing initial investigations on the data to gain **insights**, spot **anomalies**, test **hypotheses**, and **check assumptions**.

EDA typically employs **statistical graphics** and **data visualization** methods to visually represent the data.

Key points about exploratory data analysis:

1. **Purpose:** EDA is conducted to understand the data and gather insights before diving into more advanced analysis or modeling techniques.
2. **Data Exploration:** EDA involves exploring the data to identify **patterns, trends, outliers, and other features** that may be unexpected.
3. **Summary Statistics:** EDA often utilizes summary statistics, such as mean, median, mode, and skewness, to describe the central tendency and distribution of the data.
4. **Graphical Analysis:** Graphical representations, such as histograms, box plots, scatter plots, and Q-Q plots, are commonly used in EDA to visualize the data and identify relationships between variables.
5. **Data Cleaning:** EDA may also involve data cleaning and transformation processes to ensure the data is in a suitable format for analysis.

Some common data problems that can be revealed during EDA:

1. **Errors in the data:** EDA can help identify errors in the data, such as incorrect values, missing values, or inconsistencies.
2. **Outliers:** EDA can detect outliers, which are data points that significantly deviate from the rest of the data set.
3. **Unexpected patterns or trends:** EDA can uncover patterns or trends in the data that may be unexpected or contrary to initial assumptions.
4. **Variable relationships:** EDA can reveal relationships between variables, such as correlations or dependencies, which can provide insights into the data.
5. **Data inconsistencies:** EDA can identify inconsistencies in the data, such as duplicate records or conflicting information.

By conducting EDA, data practitioners can gain a better understanding of the data and address these data problems before proceeding with further analysis or modeling. This helps ensure the accuracy and reliability of the results obtained from the data.

